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GitHub Link	https://github.com/subantapoudel01/FPL-Predictor.git
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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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Catalan FPL Predictor`

1. Introduction

1.1. The Problem Domain

Fantasy Premier League (FPL) is a strategic fantasy sports, with over more than 11 million players. FPL is the biggest Fantasy football game in the world. Which is free to play and where players can win great prizes! But successful FPL manager requires a great knowledge of prediction. Where the managers predict which players might perform well in the upcoming matches. But human decision-making is often flawed, mostly because of cognitive biases such as recency bias or club bias. Often novice manager overvalues a player's last game and prefers players from supported teams. Where this project aims to address the limitations of human emotions and develop an Artificial Intelligence system that predicts players performance objectively using historical data rather than emotions and any kind of bias.

1.2. Project Aim

The main aim of the project is to create a Multiple Linear Regression model capable of calculating the specific Total Points of a player will score in the next Game week. This system helps FPL managers to identify high-value players based on statistics (Form) rather than insights.

1.3. AI Agent Definition (PEAS Framework)

“An AI system can also be called a Rational Agent. It is an entity that takes its environment in consideration through sensors and acts upon that environment through actuators that maximize a performance measurement.” (PRENTICE HALL SERIES, 2010)

The design of this FPL prediction project is to be a Goal-Based Agent. The main goal is to accurately predict the total points for each and specific players in the upcoming game week. Which had been defined by using the standard PEAS framework:

i. Performance Measure(P)

For the regression agent, the success has been determined by how close the predicted points are to the actual points achieved by the players. Which defines the criteria for the success.

- **Metric:** Mean Squared Error (RMSE)
- **Rationality:** The coefficient (weights) that mathematically minimizes the error term. It does not guess randomly, rather optimizes on SSE (Sum of Square Errors) to achieve the highest possible accuracy given the historical data.

ii. Environment (E)

The environment is the domain in which the AI agent operates, based on specific properties the environments can be classified. (PRENTICE HALL SERIES, 2010) Similarly, this project's environment which is the FPL Dataset is classified as:

- **Fully Observable:** The agent has the complete access of the data (all columns like ICT Index, Minutes, and Goals). There are no hidden information during the training process.
- **Stochastic:** Unlike Chess football is not deterministic. A player might have a high ICT Index but still fail to score or assist, due a better performance of opponents like a goalkeeper making a heroic save or a defender's goal line clearance or may be the opponent was more organized as a team rather than relying on a single player or the moment of magic in the game. In football luck is not that exists but rather created by the players from their perseverance. The agent must handle this kind of uncertainty.
- **Static:** the historical training data from season 2021 to 2025 does not change while the agent is in the training phase. The past stats are fixed during the analysis.
- **Discrete:** The games are played in discrete jumps rather than smoothly evolving.
- **Single-Agent:** To Predict the score, this model has to operate alone.

iii. Actuators (A)

The mechanisms by which the agent acts on the environment

- **Primary Actuator:** The Prediction Output Interface.
- **Action:** The output is generated when the system generates a continuous numerical value like the Predicted Point. Which helps the FPL manager's decision making.

iv. Sensors (S)

This is how the agent perceives the environment.

- **Principles:** The agent takes in the world through Data Ingestion.
- **Mechanism:** Using the Pandas library, the agent reads the .csv file, then senses the features such as ICT Form, Fixture Difficulty Rating, and value, and convert them into vector inputs for the regression equation.

2. Background

2.1. Dataset Description

The data is sourced from the Vaastav Anand Fantasy Premier League Repository on GitHub, widely the considered gold standard FPL research.

- **Volume:** Dataset contains data for over 700 players, every game week per season.
- **Scope:** For the insurances of tactical relevance, the data has been filtered to only include the post-pandemic era, which includes approximately 75000 rows of training data.
- **Key Features:** The primary features which has been included in the analysis are Minutes Played, and the ICT Index, which was the metric developed by the Premier League to qualify each player's impact on a match.

2.2. Theoretical Model: Multiple Linear Regression

Multiple Linear Regression has been chosen as the core algorithm. Designed to predict continuous variables (Points), unlike other classification modes that predict categorical outcomes a win/loss.

The linear relationship between the independent input variables (X) and the dependent target variable (y), is the mathematical model. The equation is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \varepsilon$$

Where:

- y is the predicted Total Points.
- x_1, x_2 are the input features of ICT Index, Form.
- β represents the learned weights for each feature.
- ε is the error term (residuals).

2.3. The Learning Process (Sum of Squared Errors)

By minimizing the error between its predictions and the actual results, is how the model learns the optimal weights. Which is achieved by minimizing the Sum of Squared Errors (SSE).

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Mathematical Explanation:

- $\sum$ (Sigma): Summation, which means the algorithm must iterate through every data points in the training set, from player 1 ($i = 1$) to the last (n)
- y_i : The Actual Points achieved by the players.
- $\hat{y}_i$: The Predicted Points generated.
- **Squared Term:** The difference is not squared to ensure that negative errors do not cancel out positive errors, and to heavily penalize large outliers.

2.4. Literature Review (Alternative Models)

Other algorithms were also considered but rejected:

- Naïve Bayes: In football all features are highly correlated (a player with higher number of Goals/Assists will always have a high ‘Threat’ score), which violates the naïve bayes rules, which assumes that all features are independent. (Lecture w. 5.)
- Logistic Regression: Is a classification algorithm used for binary outcomes (will a player score? Yes/No). Since the aim is to predict a precise numerical score, Logistic Regression is not applicable. (Lecture)

3. Solution

3.1. Explanation of the AI Algorithm

Utilizing a Hybrid Architecture which processes data into two different distinct stages:

First Stage: Linear Regression (Statistical Base), The core model has been trained on 75000+ historical rows, which predicts a player’s “Raw Potential” based on:

$$y = \beta_0 + \beta_1(\text{Form}) + \beta_2(\text{ICT Index}) + \beta_3(\text{Value}) + \varepsilon$$

Which captures the statistical baseline of the player.

Second Stage: Expert's Rules, the tactical nuances of football cannot be understood just by Pure regression. The application applies a post-processing logic layer:

- a. The “Big Club Tax”: Attackers who are cheap ($< 9.0m$) and play for elite clubs (man city, arsenal) to handle rotations of the players.
- b. Fixture Difficulty Adjustments:
 - Easy Match: * 1.35 boost.
 - Hard Match: * 0.70 penalty.
- c. Strict Starter Probability: For the “Benchwarmer” issue, Penalizing the players averaging < 45 minutes, reducing their expected points to near zero.

3.2. Pseudocode of the Solution

The logic within the system, illustrating the *make_predictions*.

```

FUNCTION make_predictions(player_data, model):
  FOR EACH player IN player_data:
    # Step 1: Statistical Prediction
    input_vector = [Form, ICT_Index, Minutes_Avg, Price]
    raw_xp = model.predict(input_vector)
    # Step 2: Expert Rule Application
    # Rule A: Starter Probability Curve
    IF average_minutes > 60:
      starter_prob = 1.0
    ELSE IF average_minutes < 45:
      starter_prob = 0.25 # High risk of benching
    # Rule B: Fixture Difficulty Multiplier
    difficulty = get_opponent_difficulty(player.next_match)
    IF difficulty == 1 (Easy):
      fixture_multiplier = 1.35
    ELSE IF difficulty == 5 (Hard):
      fixture_multiplier = 0.70

```



```

# Rule C: The "Big Club Tax" (Bias Correction)

rotation_penalty = 1.0

IF player.team IN ["MCI", "ARS", "LIV"] AND player.price < 9.0:

    rotation_penalty = 0.75 # 25% penalty for non-premium assets

# Step 3: Final Calculation

final_prediction = raw_xp * fixture_multiplier * rotation_penalty * starter_prob

# Step 4: Safety Clamp

final_prediction = CLAMP(final_prediction, MIN=0, MAX=18)

RETURN sorted_predictions

END FUNCTION

```

3.3. Diagrammatical Representation

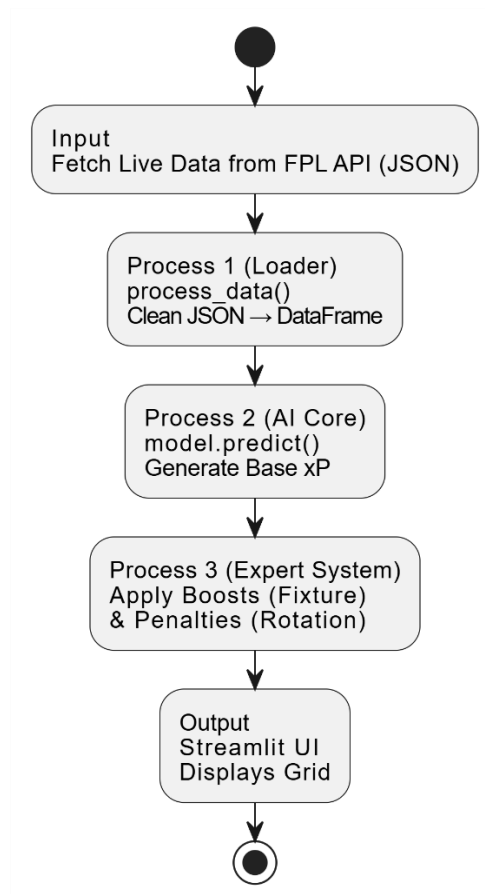


Figure 1: System Flowchart illustrating the Hybrid Architecture pipeline.

3.4. Development Process

Using an Agile Methodology, which allowed for iterative improvements, based on the testing results.

Tools & Technologies:

- Python:** The Core language for programming
- Pandas:** For complex data manipulation
- Scikit-Learn:** Training the Linear Regression Model.
- Streamlit:** For building the interactive interface
- Joblib:** For creating the .pkl from the training model for instant loading.
- VS Code:** Integrated Development Environment.

3.5. Achieved Results

Successfully achieved live, interactive predictions.

- Accuracy: For the Game week 22 the model was able to achieve an RMSE of ~ 1.94 .
- Validation: The “Backtest” feature was correctly predicting low scores for rotated players like Rico Lewis.

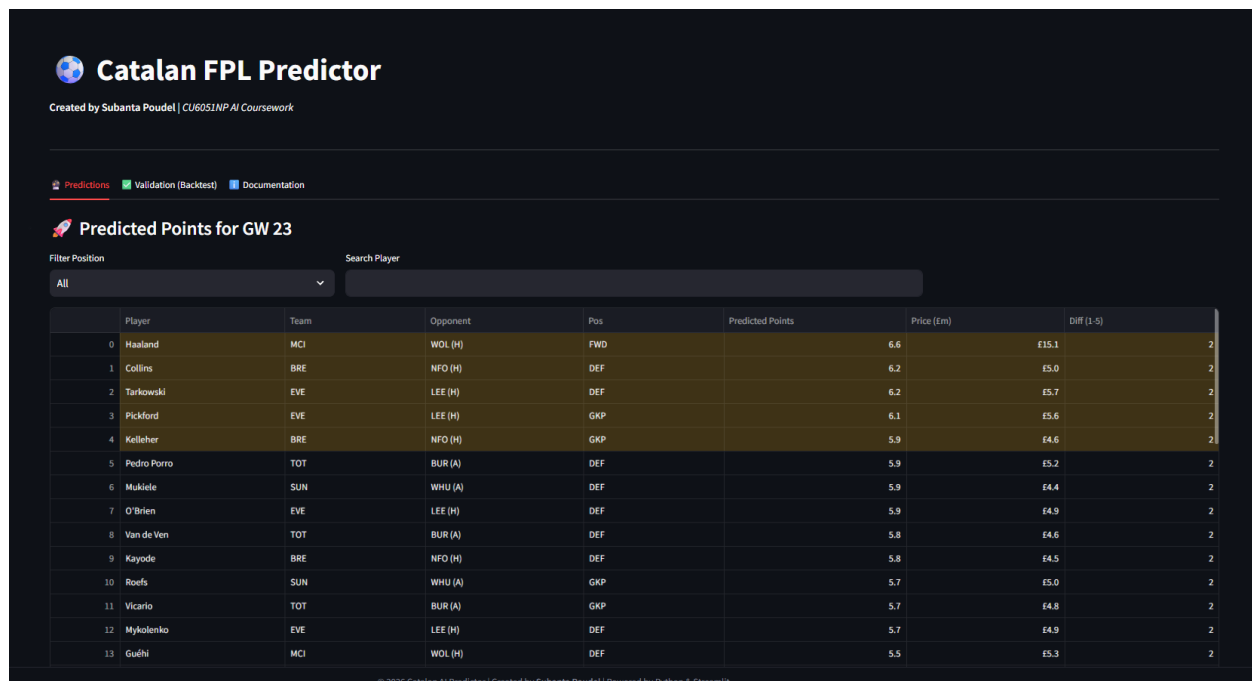


Figure 2: The Main Prediction Dashboard showing top 5 Recommended Captains.

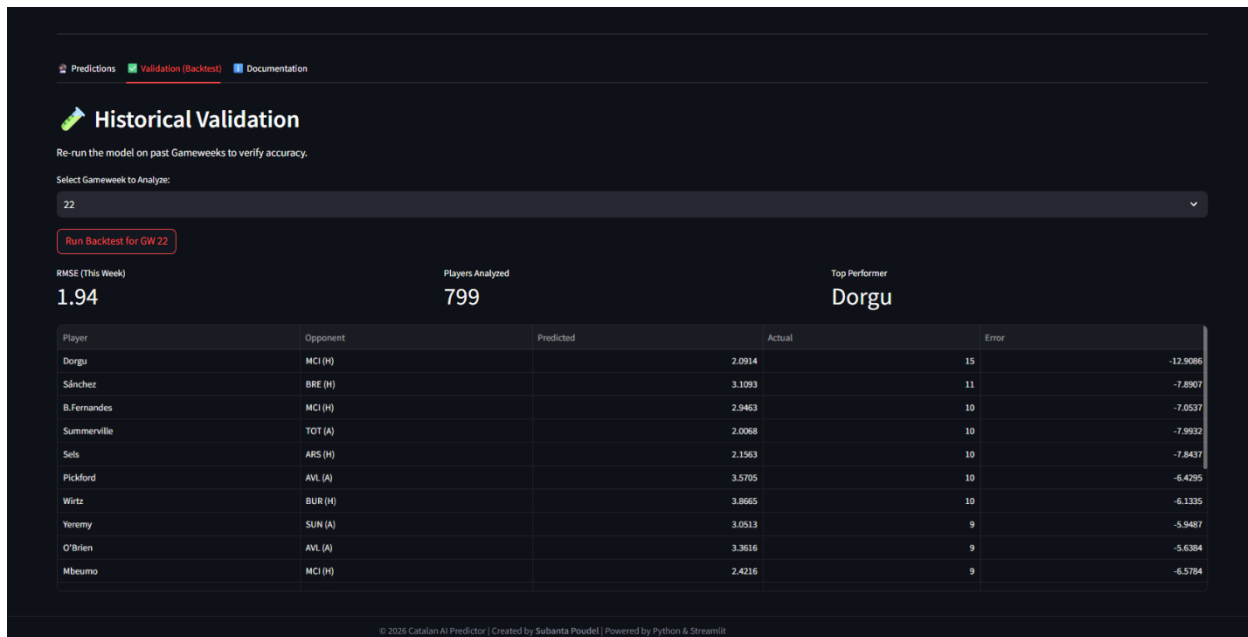


Figure 3: The Historical Validation Interface allowing users to test accuracy on past gameweeks.

4. Conclusion

4.1. Analysis of Work Done

The project successfully was able to met the primary aim of the creating an objective prediction tool. Transitioning form the pure Linear Regression model to a Hybrid Architecture which was the critical success factor. The Linear Regression only model did not factor the complex tactical nature of football, Like even if a player is 100% fit they might not be playing, they might not start the game, they might be on the bench. By using the Expert's Rules on top of the AI, the smart was able to achieve a balance between statistical power and tactical reality.

4.2. Real World Application

Addressing the Cognitive Bias problem. And can be used on other systems not only football only. This same architecture can also be applied to sales forecasting, risk scoring, or performance prediction in other domains. Although Fantasy Premier League data is used to train this AI system, its basic architecture—a Hybrid Intelligence System that combines statistical regression with rule-based domain constraints—directly reflects how artificial intelligence is used in crucial enterprise industries:

- **Sales & Demand Forecasting:** In supply chain management and retail, abrupt market disruptions or extreme weather occurrences are frequently overlooked by pure statistical regression models. Like how our FPL model uses a "Big Club Tax" and a "Fixture Difficulty Multiplier" to modify raw statistical point forecasts, retail AI systems overlay business heuristic rules on past rolling sales averages to avoid stockouts or overstocking in erratic market conditions.
- **Financial Risk Scoring:** Similar two-stage pipelines are used by banks and credit organizations to assess loan eligibility. Strict, rule-based regulatory "circuit breakers" (like our starter probability thresholds) override the AI to automatically reject applicants who fall below mandatory compliance limits, regardless of their raw statistical score. A foundational machine learning model determines an applicant's continuous creditworthiness score based on historical transaction form.

Healthcare Triage & Resource Allocation: Hospital bed triage is intimately correlated with player availability and point returns. Hospital AI algorithms classify patients into priority tiers based on both continuous patient vital signs and discrete risk criteria (such as age or unexpected injury). The mathematical reasoning behind our step-curve starter probability heuristic, which lowers predicted points for players averaging less than 45 minutes, is identical to that used to de-escalate lower-risk patients from intensive care units.

4.3. Further Work

Further iterations to improve the prediction accuracy:

- **Non-Linear Models:** Implement LST to capture sequential from trends compared to simple rolling averages.
- **Fixture Congestion Analysis:** Players have to play in multiple competitions which are cramped short period of time, so integrating rest days feature to improve rotation prediction.
- **Sentiment Analysis:** Using NLP to get the injury news from the press conferences, not in the official API.
- **Backtest Refinement:** implement a robust historical injury database would allow for perfect historical simulation.

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